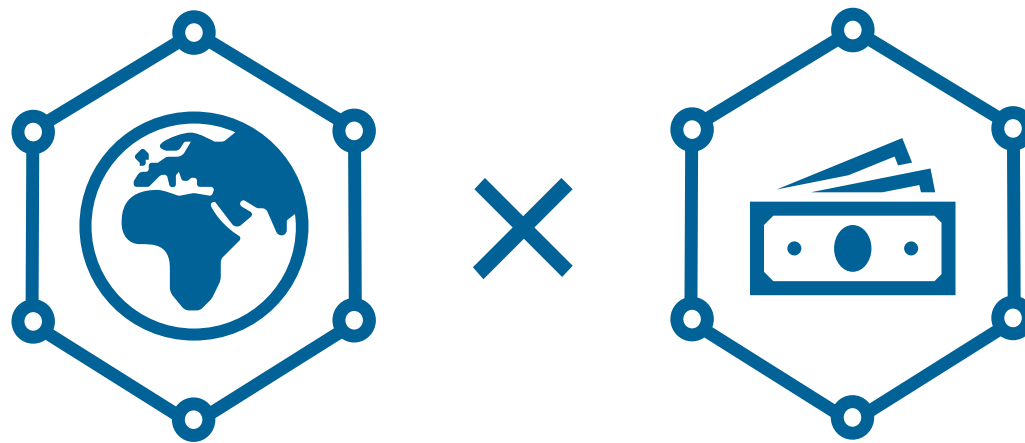




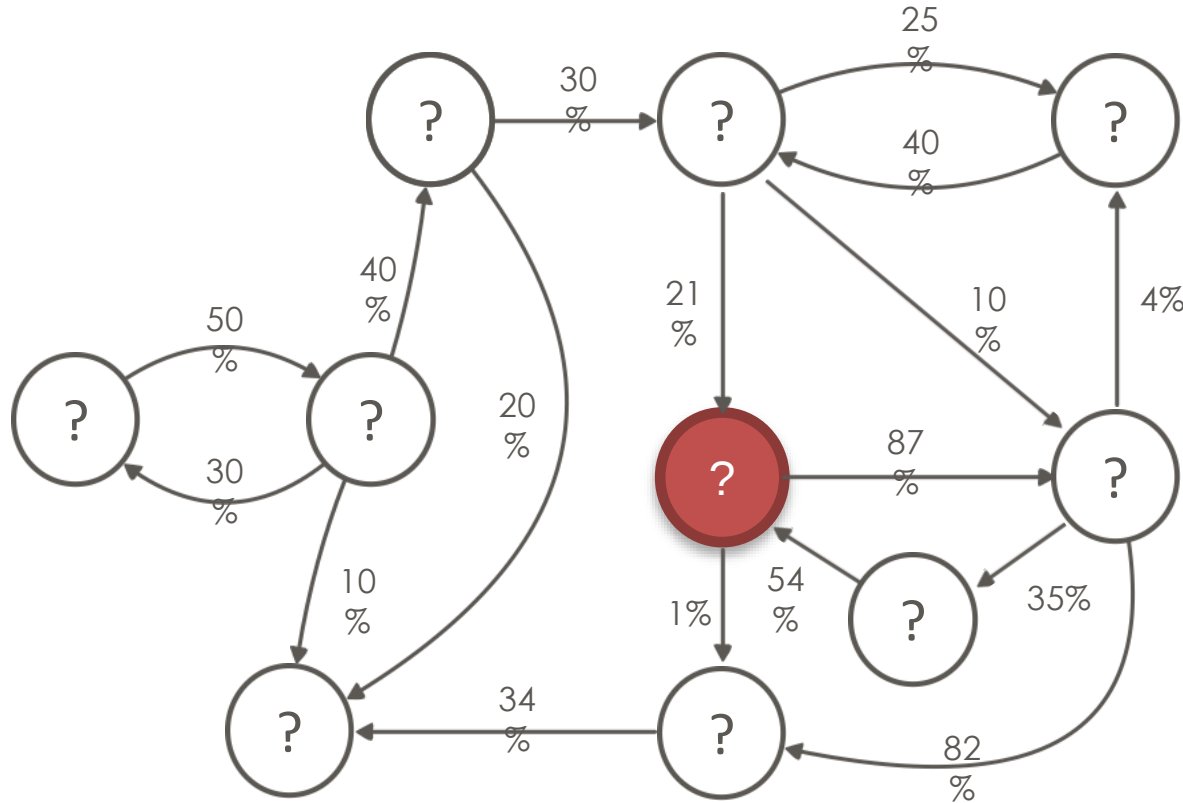
Financial Knowledge Graphs

# Company Ownership and Control

Emanuel Sallinger



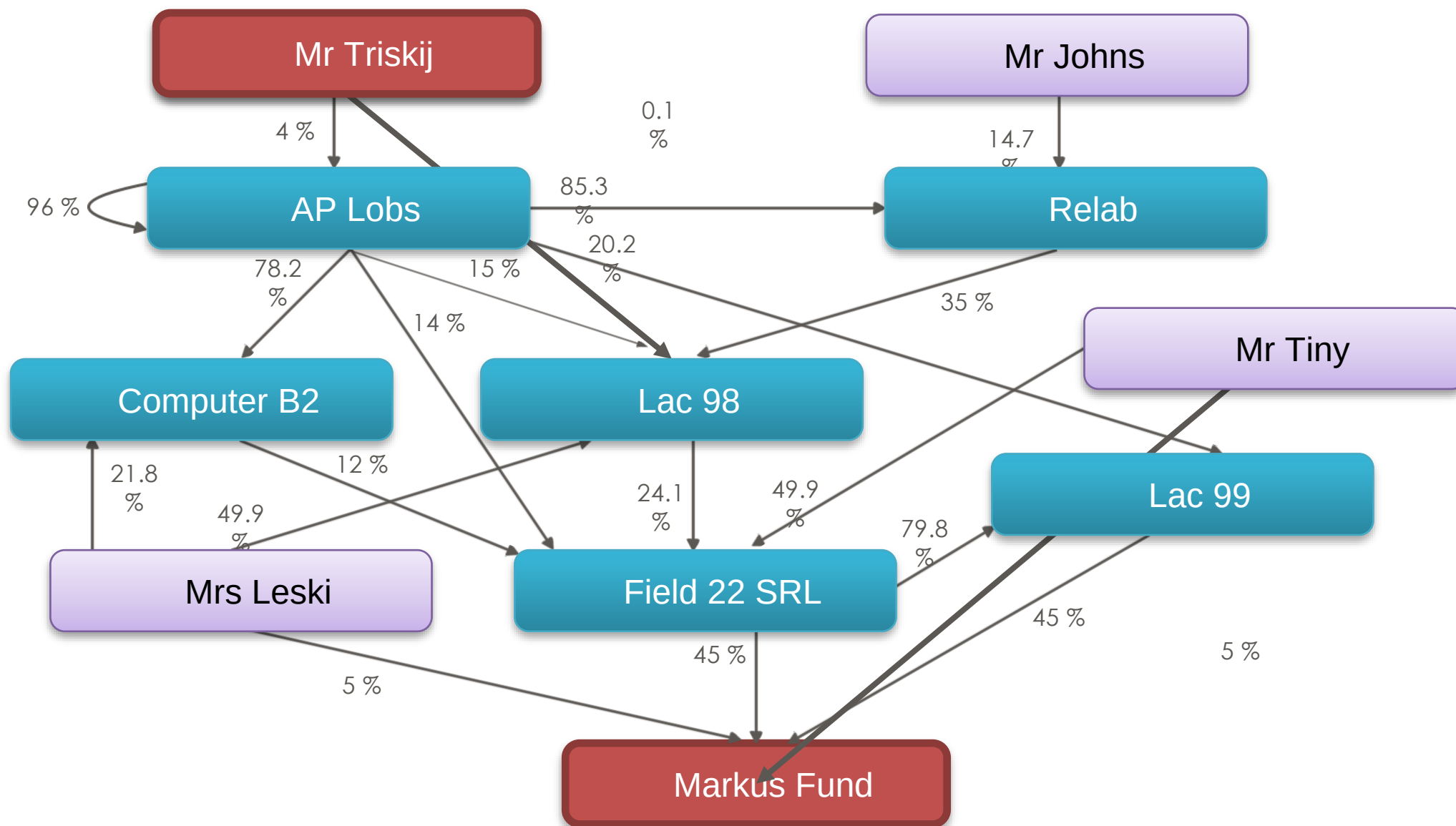
# The Setting



- Who takes **decisions**?
- Who's the ultimate **beneficial owner**?
- Is there **collusion**?
- How does **risk** propagate?
- What are the **real cash flows**?



# The Setting





# Ownership and Control



- **Integrated ownership** is about direct and indirect, owners of a company
  - it can be seen in terms of cash flow rights

- **Control** is about voting power
  - of any direct and indirect owner of a company





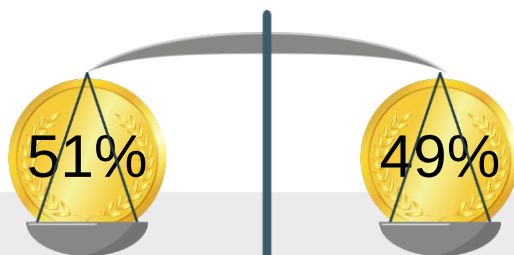
# Ownership and Control

Companies A and B have both **direct** (shares) and **indirect** ownership of company C.

A has 51% of C

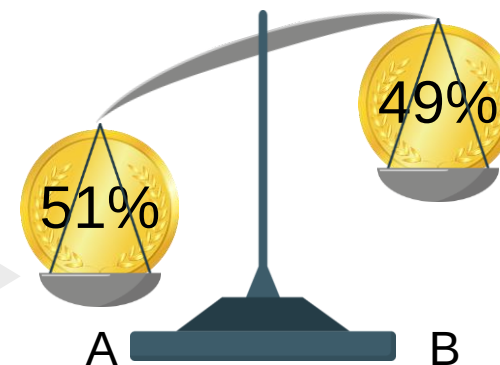
B has 49% of C

## Ownership



Ownership of C is comparable

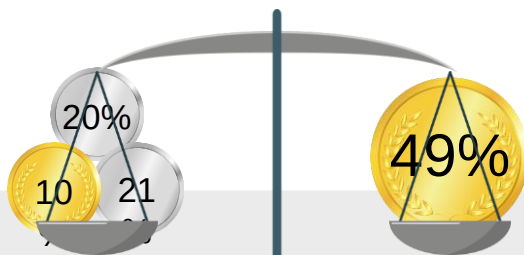
## Control



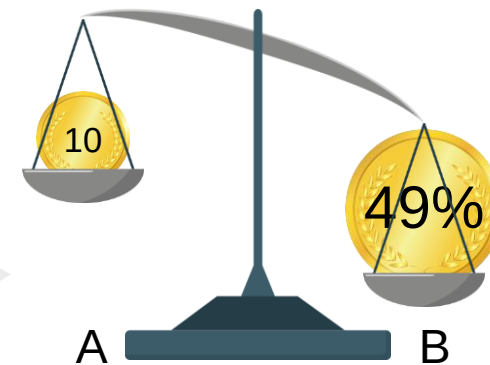
A has 100% decision power  
**A controls C**

A has 10% and 41% of C

B has 49% of C



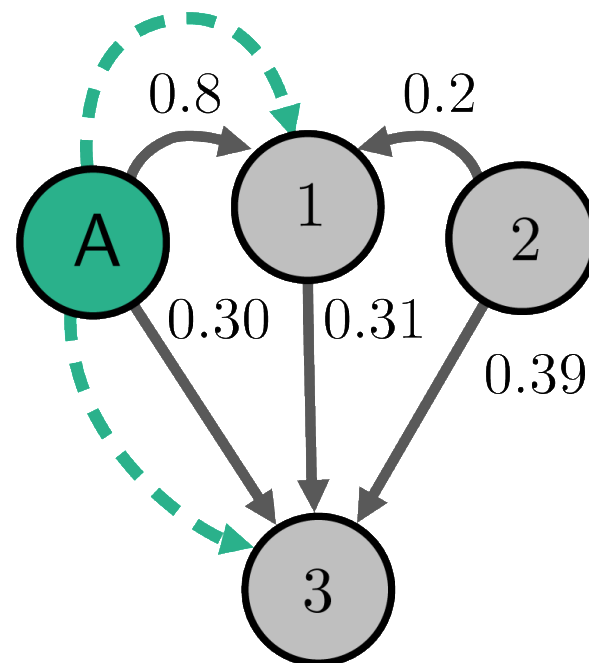
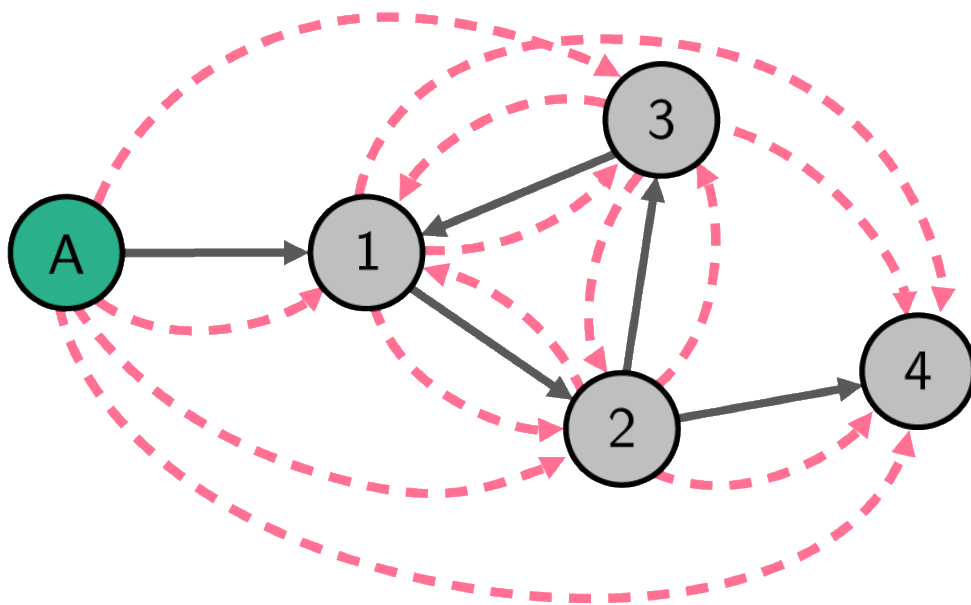
Ownership of C is comparable



A has 10% decision power  
**A does not control C**



# Integrated Ownership vs Company Control

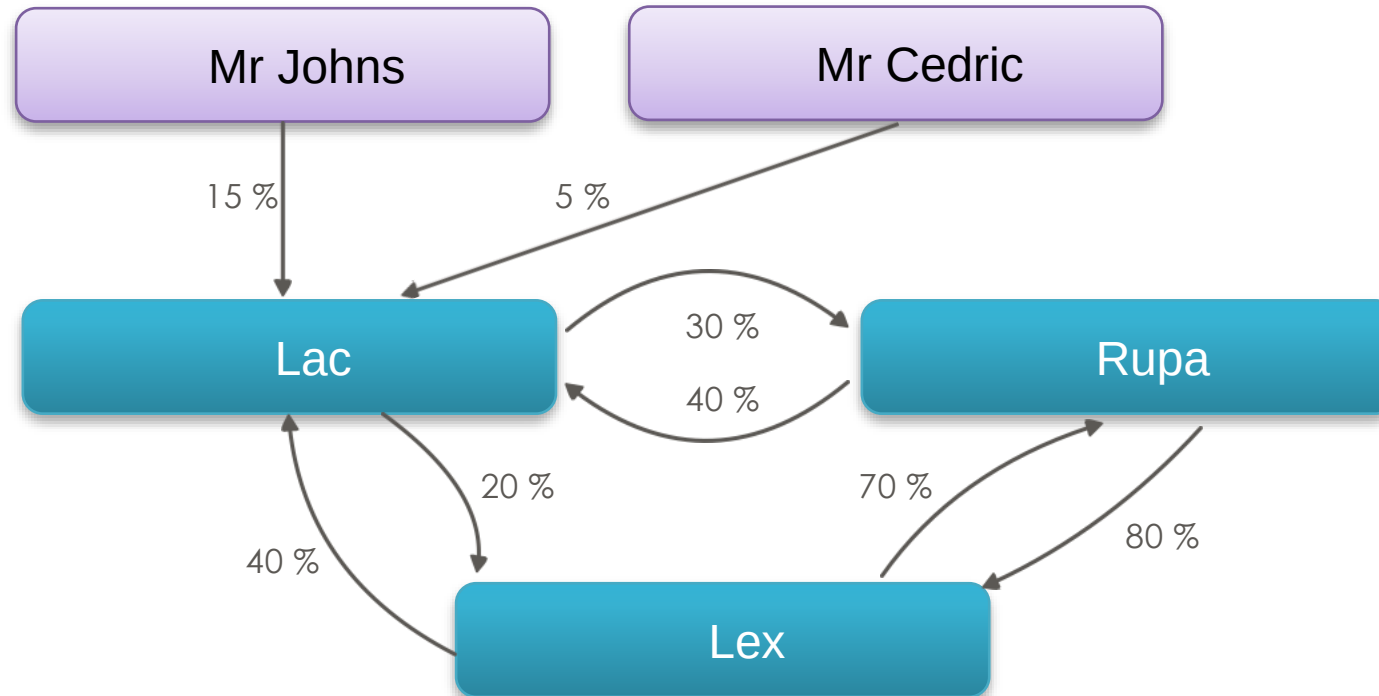




# Integrated Ownership



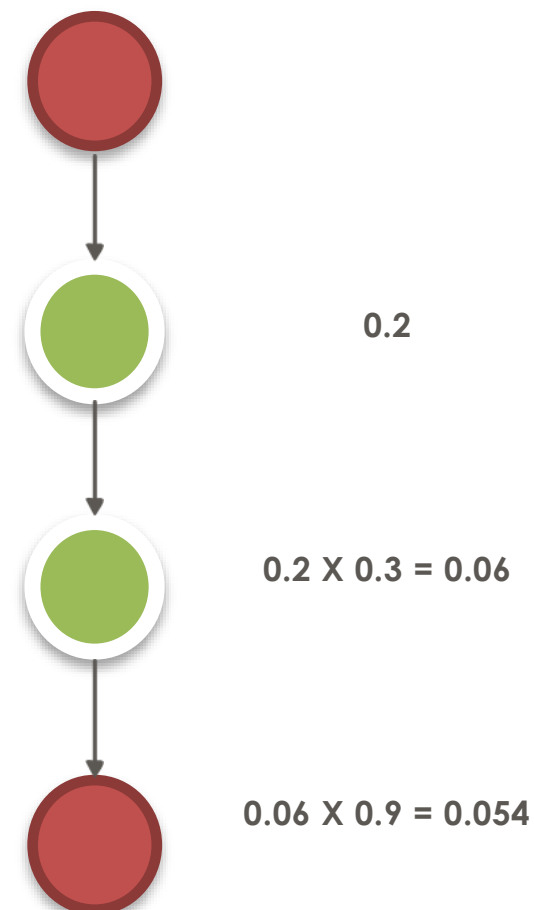
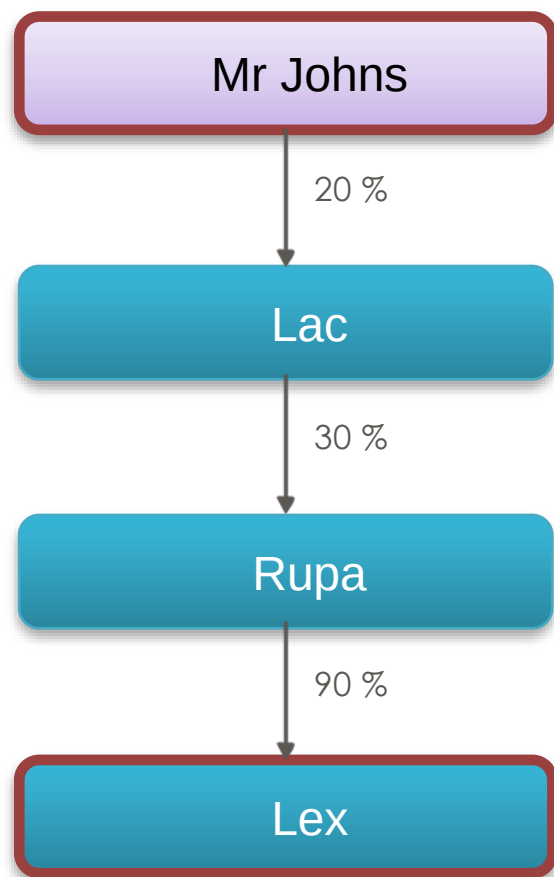
- Traditional systems only store the first-level (the closest) shareholders for a specific company





# Integrated Ownership: The Basic Math

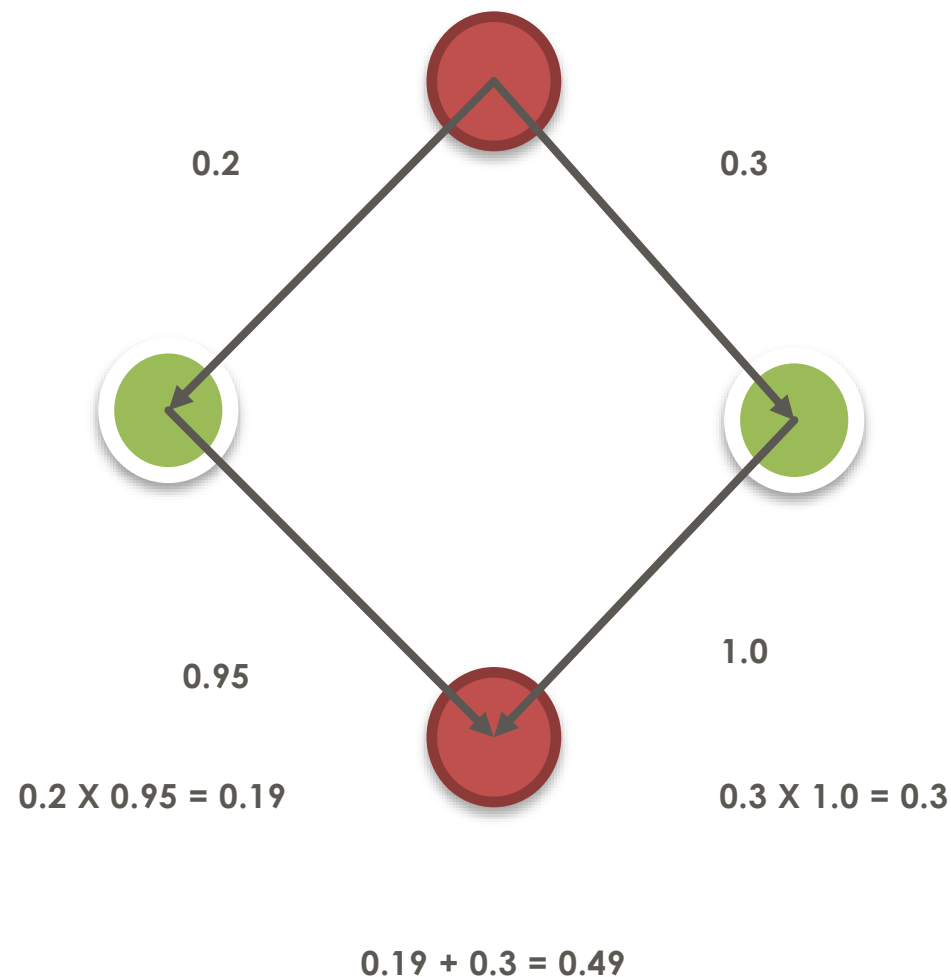
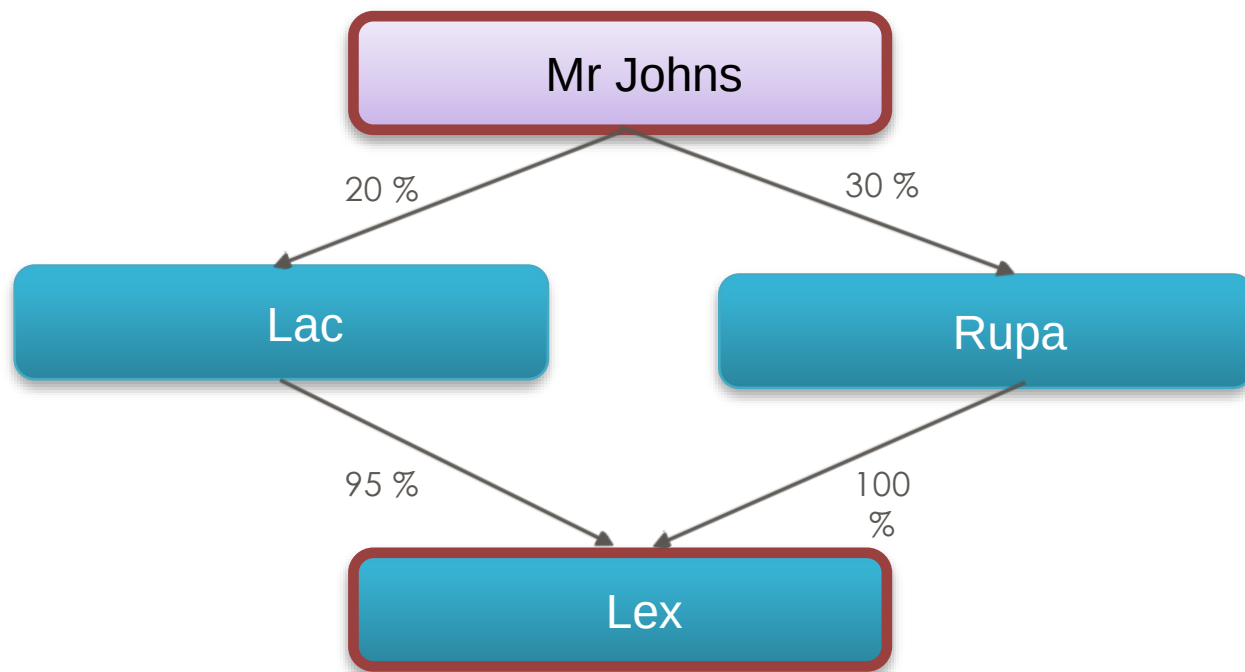
- Indirect ownership





# Integrated Ownership: The Basic Math

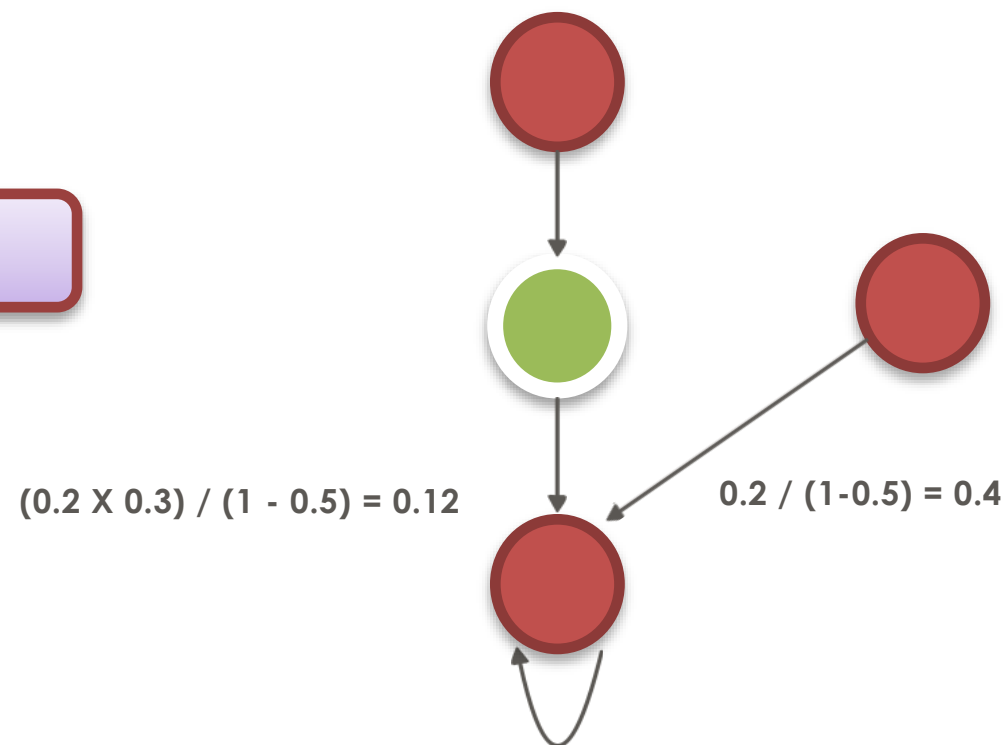
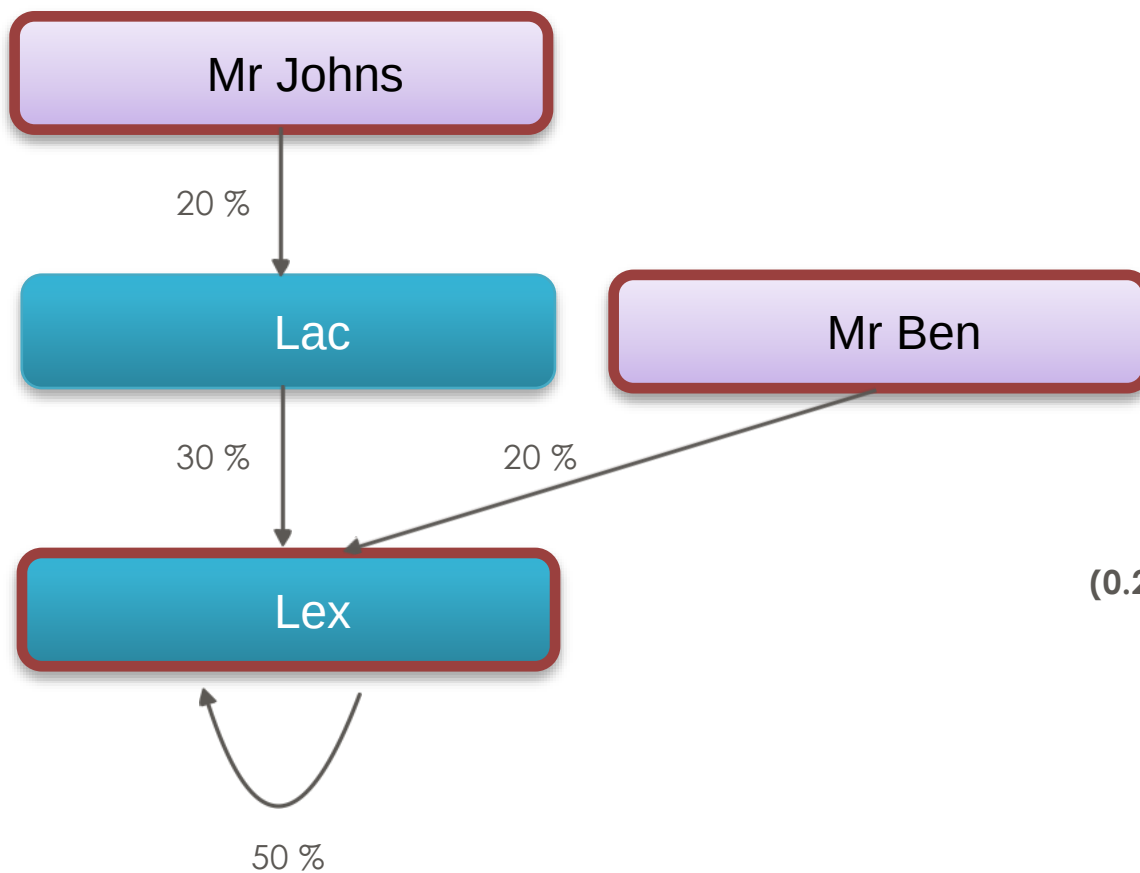
- Parallel ownership





# Integrated Ownership: The Basic Math

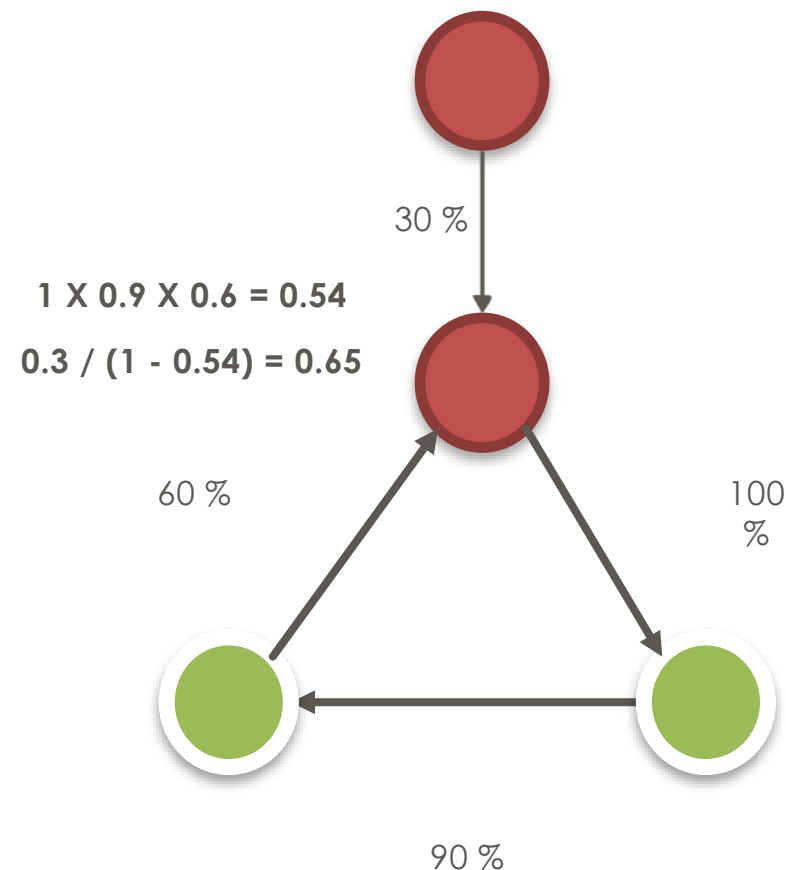
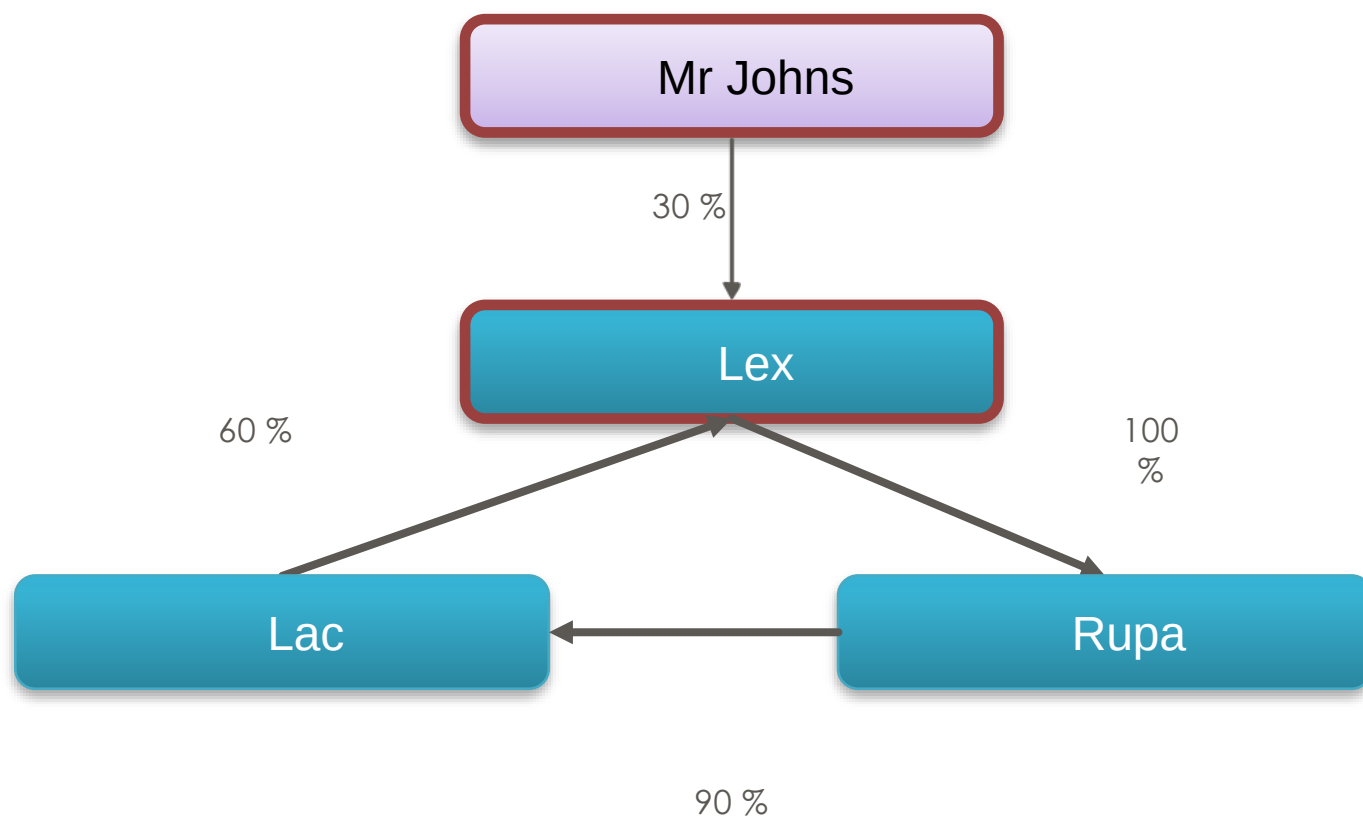
- Cycles (direct)





# Integrated Ownership: The Basic Math

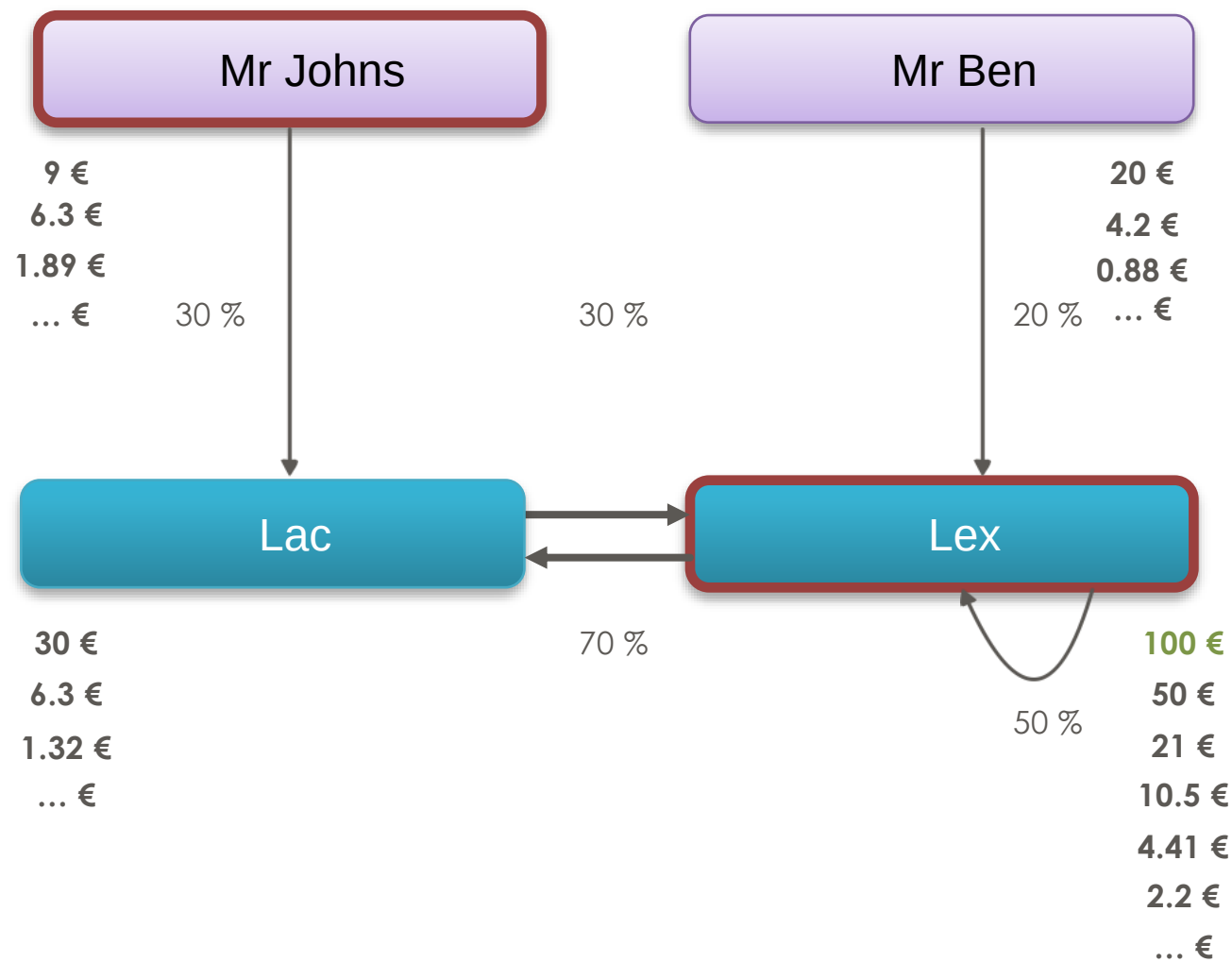
- Cycles (indirect)





# Integrated Ownership: The Basic Math

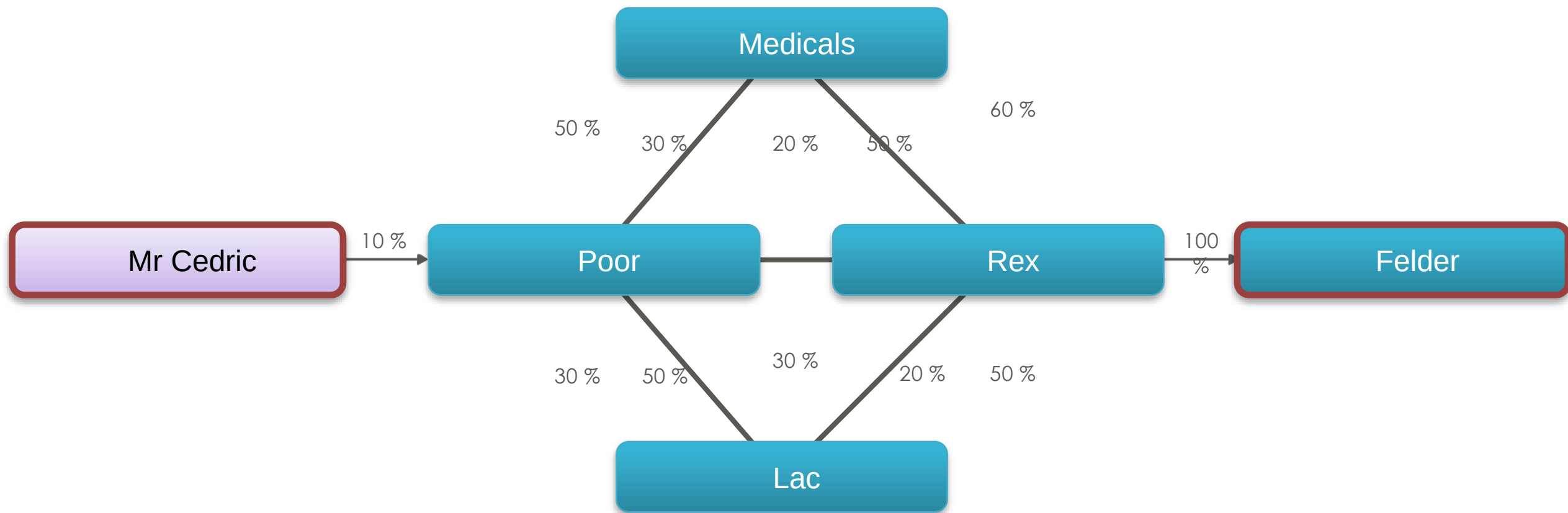
- Cycles (nested)





# Integrated Ownership: The Basic Math

- Cycles (nested)



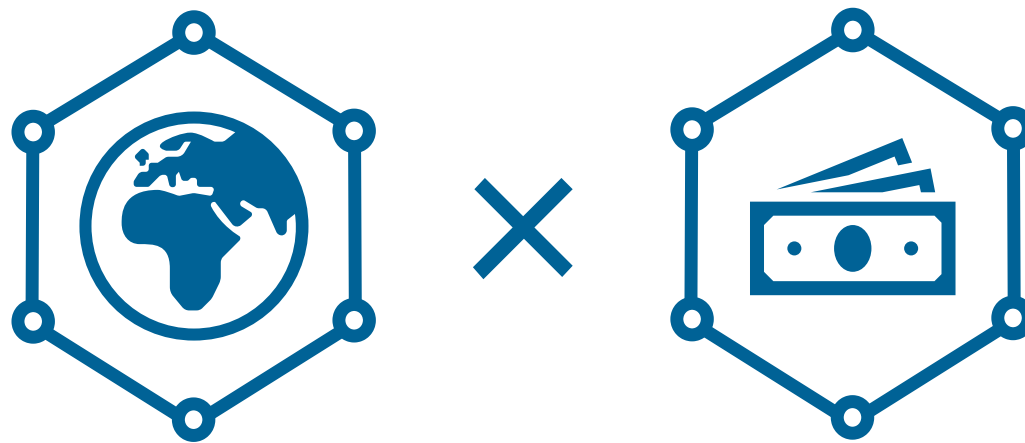


# Integrated Ownership: Computation Methods

- Problem complexity

*Computing “all-to-all” integrated (Baldone) ownerships can be solved in polynomial time in the number of companies. Conjecture:  $n^y, y \in [2,3]$*

- Our approaches:
  - **Closed-form expression** (exact solution, but unfeasible with large —real— graphs)
  - **Pure KG reasoning** (approximate solution, efficient results)
  - **Ad-hoc algorithm** (exact solution, top-level performance, ad-hoc projects)
  - **Closed-form + KG reasoning** (exact and efficient results, reusable and explainable)



Financial Knowledge Graphs

# Company Ownership and Control

Emanuel Sallinger